

Fall 2025 DRP Presentation

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Motivation

When working in the ring $\mathbb{Z}/p\mathbb{Z}$ one natural question one might have would be 'for some $a \in \mathbb{Z}/p\mathbb{Z}$ is a a square'? Or, more formally,

For some $a \in \mathbb{Z}/p\mathbb{Z}$ does there exist a solution to the congruence

$$x^2 \equiv a \pmod{p}$$

For small primes, this is an easy question as we can simply use direct calculation. For example, if I wanted to know if 2 was a square mod 7 all I have to do check the squares of $0, 1, 2, \dots, 6 \pmod{7}$. Very quickly we can see that $3^2 = 9 \equiv 2 \pmod{7}$ and so 2 is in fact a square mod 7. But what if I wanted to know if 13 was a square mod 73? This is the question I will attempt to answer today by proving the law of quadratic reciprocity.

Background (The Ring of Integers Modulo a Prime)

As I mentioned earlier we will spend a lot of time working in the ring $\mathbb{Z}/n\mathbb{Z}$ which is the integers modulo n . It's important that we have a good grasp on this ring before moving forward.

Definition

Let n be an integer. Then the ring of integers modulo n , denoted $\mathbb{Z}/n\mathbb{Z}$ is the ring with operations addition and multiplication, and whose elements are the equivalence classes modulo n .

$$\mathbb{Z}/n\mathbb{Z} = \{0, 1, 2, \dots, n-1\}$$

It is fairly easy to see that $\mathbb{Z}/n\mathbb{Z}$ forms a ring (with addition and multiplication defined as usual).

Background (Algebraic Integers)

Definition

An **algebraic integer** is a complex number ω that is the root of a polynomial $x^n + a_1x^{n-1} + \cdots + a_n = 0$, where $a_1, a_2, \dots, a_n \in \mathbb{Z}$.

Theorem

The set of algebraic integers forms a ring.

When working in the ring of algebraic integers, Ω , we define congruence as follows. If $\omega_1, \omega_2, \gamma, \alpha \in \Omega$, we say that $\omega_1 \equiv \omega_2 \pmod{\gamma}$ if $\omega_1 - \omega_2 = \gamma\alpha$. Congruence in Ω follows all the standard properties that hold in $\mathbb{Z}$. We also have the following that is true in Ω : If $\omega_1, \omega_2 \in \Omega$ and $p \in \mathbb{Z}$ is a prime, then

$$(\omega_1 + \omega_2)^p \equiv \omega_1^p + \omega_2^p \pmod{p}.$$

This fact can be easily proven by expansion with the binomial theorem and the fact that $p \mid \binom{p}{k}$ for $1 \leq k \leq p-1$.

Background (Legendre Symbol)

Definition

Let p be a prime and a any integer. We then define

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a square mod } p \\ -1 & \text{if } a \text{ is not a square mod } p \\ 0 & \text{if } p \mid a \end{cases}$$

as the **Legendre Symbol**.

Properties of the Legendre Symbol

- ❶ $a^{p-1}/2 \equiv (a/p) \pmod{p}$
- ❷ $(ab/p) = (a/p)(b/p)$
- ❸ If $a \equiv b \pmod{p}$ then $(a/p) = (b/p)$

The Law of Quadratic Reciprocity

With the following background established we can now state the Law of Quadratic Reciprocity:

Theorem

For two odd primes p and q with $p \neq q$

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\left(\frac{p-1}{2}\right)\left(\frac{q-1}{2}\right)}$$

We can notice that for any odd number, n , $n \equiv 1$ or $n \equiv 3 \pmod{4}$. If $n \equiv 1 \pmod{4}$ then $n - 2/2$ will be an even quantity. If $n \equiv 3 \pmod{4}$ then $n - 2/2$ will be an odd quantity. With this in mind we restate the theorem as follows: if **either** $p \equiv 1$ or $q \equiv 1 \pmod{4}$ then $(p/q) = (q/p)$, but if **both** $p \equiv 3$ and $q \equiv 3 \pmod{4}$ then $(p/q) = -(q/p)$. We also have the following 'baseline' values:

$$(-1/p) = (-1)^{(p-1)/2}$$

$$(2/p) = (-1)^{(p^2-1)/8}$$

Foundations of the Proof (Roots of Unity)

Definition

An n -th **root of unity** is any value that is a solution to the equation

$$x^n - 1 = 0$$

For the rest of this discussion we shall let $\zeta = e^{2\pi i/p}$ which is a p -th root of unity. The following lemma is an important fact about ζ that we will use in our proof.

Lemma

The sum $\sum_{t=0}^{p-1} \zeta^{at}$ equals p if $a \equiv 0 \pmod{p}$ and is 0 otherwise.

Proof.

If $a \equiv 0 \pmod{p}$ then $\zeta^a = 1$ by definition of a root of unity. It follows that $\sum_{t=0}^{p-1} \zeta^{at} = p$. If $a \not\equiv 0 \pmod{p}$ then $\zeta^a \neq 1$ and $\sum_{t=0}^{p-1} \zeta^{at} = (\zeta^{ap} - 1)(\zeta^a - 1) = 0$. □

Foundations of the Proof (Quadratic Gauss Sums)

Corollary to Lemma

$p^{-1} \sum_{t=0}^{p-1} \zeta^{t(x-y)} = \delta(x, y)$, where $\delta(x, y) = 1$ if $x \equiv y \pmod{p}$ and $\delta(x, y) = 0$ if $x \not\equiv y \pmod{p}$.

Definition

A quadratic gauss sum, g_a is defined as

$$g_a = \sum_{t=0}^{p-1} \left(\frac{t}{p} \right) \zeta^{at}$$

with g_1 commonly notated as g .

It is important to note that $g_a = (a/p)g$.

Lemma

$$g^2 = (-1)^{(p-1)/2} p$$

Foundations of the Proof (Quadratic Gauss Sums) (cont.)

To prove the lemma we will consider the sum $\sum_a^{p-1} g_a g_{-a}$ in two different ways and our result will follow from their equality. First, consider that if $a \not\equiv 0 \pmod{p}$ then $g_a g_{-a} = (a/p)(-a/p)g^2$ by the fact on the previous slide. However, we can simplify this further,

$$(a/p)(-a/p)g^2 = (a/p)(a/p)(-1/p)g^2 = (a^2/p)(-1/p)g^2 = (-1/p)g^2$$

by repeated application of the multiplicative property of the Legendre symbol and the fact that $(a^2/p) = 1$ since a^2 will (trivially) be a square mod p . Now, revisiting our sum:

$$\sum_{a=0}^{p-1} g_a g_{-a} = \left(\frac{-1}{p}\right) (p-1) g^2.$$

Note that despite the fact we are summing over p terms, we have a $(p-1)$ in our closed form since $a \not\equiv 0 \pmod{p}$.

Foundations of the Proof (Quadratic Gauss Sums) (cont.)

The second way we will evaluate the sum is by definition of a Gauss sum. Using the definition we have

$$g_a g_{-a} = \sum_{x=0}^{p-1} \sum_{y=0}^{p-1} \left(\frac{x}{p} \right) \left(\frac{y}{p} \right) \zeta^{a(x-y)}.$$

Now, we sum the entire thing over a :

$$\begin{aligned} \sum_{a=0}^{p-1} g_a g_{-a} &= \sum_a \sum_x \sum_y \left(\frac{x}{p} \right) \left(\frac{y}{p} \right) \zeta^{a(x-y)} \\ &= \sum_x \sum_y \left(\frac{x}{p} \right) \left(\frac{y}{p} \right) \sum_a \zeta^{a(x-y)} \\ &= \sum_x \sum_y \left(\frac{x}{p} \right) \left(\frac{y}{p} \right) \delta(x, y) p \end{aligned}$$

Foundations of the Proof (Quadratic Gauss Sums) (cont.)

$$= \sum_x \sum_y \left(\frac{x}{p}\right) \left(\frac{y}{p}\right) \delta(x, y) p$$

Notice that by definition of $\delta(x, y)$, this sum will only ever be nonzero when $x \equiv y \pmod{p}$. There are only $p - 1$ pairs of x and y such that $\delta(x, y)$ is nonzero. Thus,

$$\sum_x \sum_y \left(\frac{x}{p}\right) \left(\frac{y}{p}\right) \delta(x, y) p = (p - 1) p.$$

Now combining this with our previous equality we find that

$$\left(\frac{-1}{p}\right) (p - 1) g^2 = (p - 1) p \Rightarrow g^2 = \left(\frac{-1}{p}\right) p$$

which was desired.

Proof of the Law of Quadratic Reciprocity

With this information now in mind we can start with the proof. Consider $p^* = (-1)^{(p-1)/2} p$, and another odd prime q such that $q \neq p$. We will be working with congruences mod q in the ring of algebraic integers:

$$g^{q-1} = (g^2)^{(q-1)/2} = p^{*(q-1)/2} \equiv \left(\frac{p^*}{q}\right) \pmod{q}$$
$$\Rightarrow g^q \equiv \left(\frac{p^*}{q}\right) g \pmod{q}$$

Rewriting g^q using the definition of the Gauss sum gives

$$g^q = \left(\sum \left(\frac{t}{p}\right) \zeta^t\right)^q \equiv \sum \left(\frac{t}{p}\right)^q \zeta^{qt} \equiv g_q \pmod{q}$$

Recalling that $g_a = (a/p)g$, we now have that $g^q \equiv g_q \equiv (q/p)g \pmod{q}$, thus

$$\left(\frac{q}{p}\right) g \equiv \left(\frac{p^*}{q}\right) q \pmod{q}$$

Proof (cont.)

If we multiply both sides by g and apply the fact that $g^2 = p^*$ we have

$$\begin{aligned}\left(\frac{q}{p}p^*\right)p^* &\equiv \left(\frac{p^*}{q}\right)p^* \pmod{q} \\ \Rightarrow \left(\frac{q}{p}\right) &\equiv \left(\frac{p^*}{q}\right) \pmod{q} \\ \Rightarrow \left(\frac{q}{p}\right) &= \left(\frac{p^*}{q}\right)\end{aligned}$$

Which was to be shown.

Worked Example

Earlier I asked if 13 was a square, or quadratic residue, mod 73. With the law of quadratic reciprocity we can now figure this out:

$$\begin{aligned}\left(\frac{13}{73}\right) &= (-1)^{\left(\frac{13-1}{2}\right)\left(\frac{73-1}{2}\right)} \left(\frac{73}{13}\right) \\ &= \left(\frac{8}{13}\right) = \left(\frac{-5}{13}\right) = \left(\frac{-1}{13}\right) \left(\frac{5}{13}\right) = (-1)^{\frac{5-1}{2}} \left(\frac{5}{13}\right) \\ &= (-1)^{\left(\frac{13-1}{2}\right)\left(\frac{5-1}{2}\right)} \left(\frac{13}{5}\right) = \left(\frac{3}{5}\right) = \left(\frac{-2}{5}\right) \\ &= \left(\frac{-1}{5}\right) \left(\frac{2}{5}\right) = (-1)^{(5-1)/2} (-1)^{(5^2-1)/8} = -1\end{aligned}$$

Therefore, 13 **is not** a square mod 73.